Deep Learning Systems Analysis Report: IMDB Sentiment Classification with LSTM

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GitHub Repository: https://github.com/sandipdey2/imdb-sentiment-rnn

Report Overview

This project implements a recurrent neural network for binary sentiment classification on movie reviews. The dataset is the IMDB Movie Reviews collection (50,000 labeled reviews). A single-layer LSTM was trained as the baseline, and one controlled experiment changed the learning rate while holding architecture and all other training choices fixed. The work demonstrates correct PyTorch implementation, experimental reasoning, and interpretation of training dynamics on sequential text data.

Dataset and Task Description

The IMDB dataset contains English-language movie reviews labeled positive or negative. It contains 50,000 samples and is a standard public benchmark for sequence classification. Each review is a variable-length sequence of words, making it appropriate for recurrent architectures that process tokens in order. After tokenization and padding to a fixed maximum length of 200, the data are suitable for batch training of an LSTM. The source is the Stanford AI Lab release (https://ai.stanford.edu/~amaas/data/sentiment/).

Model Architecture and Design Decisions

The baseline model consists of an embedding layer (vocabulary size 8,000 mapped to 128-dimensional dense vectors), a single-layer LSTM with hidden size 128 that maintains a hidden state across the sequence, dropout (p=0.3) applied to the final hidden state, and a linear output layer that produces a logit for binary classification. The model has 1,156,225 parameters. The embedding layer captures token semantics, the LSTM captures sequential dependencies, and dropout provides mild regularization (Hochreiter & Schmidhuber, 1997). Binary cross-entropy with logits was used as the loss, and Adam was chosen as the optimizer for its adaptive learning rates (Kingma & Ba, 2015). These choices are standard for text classification with moderate-sized vocabularies. Training used an 80/20 split (40,000 train / 10,000 validation) and ran on Apple MPS.

Experimental Comparison

The single experimental change was the learning rate: the baseline used 1e-3 while the experimental run used 5e-3. All other factors—architecture, number of layers, embedding dimension, hidden size, dropout rate, batch size, number of

epochs (40), optimizer, and data splits—remained identical. Learning rate is a major hyperparameter that directly affects step size and training stability, making it a meaningful controlled comparison (Smith, 2017).

Results and Interpretation

Both models were trained for 40 epochs and evaluated on the held-out validation set of 10,000 samples. Exact results from the notebook runs:

Baseline (lr = 1e-3):
- Epoch 1: Train Loss 0.6930 | Val Loss 0.6908
- Epoch 7: Train Loss 0.2507 | Val Loss 0.3382 (best validation loss)
- Epoch 40: Train Loss 0.0084 | Val Loss 1.0271
- Validation accuracy: 0.8535

Experiment (lr = 5e-3):
- Epoch 1: Train Loss 0.6353 | Val Loss 0.5027
- Epoch 7: Train Loss 0.2216 | Val Loss 0.3517 (near-best validation loss)
- Epoch 40: Train Loss 0.5054 | Val Loss 0.5621
- Validation accuracy: 0.7160

The baseline with the lower learning rate reached a higher final validation accuracy (0.8535 vs 0.7160). However, after approximately epoch 7 the baseline's validation loss began rising while training loss continued to fall (ending at 0.0084 train vs 1.0271 val), a clear sign of overfitting. The higher learning rate produced faster early improvement but became unstable later, with both train and validation loss increasing after the midpoint and finishing with substantially lower accuracy. One concrete observation is that the experimental run's training loss rose from 0.1666 at epoch 19 to 0.5054 at epoch 40, indicating divergence under the larger step size. These patterns illustrate the classic trade-off between convergence speed and training stability, and the importance of early stopping when training an LSTM on text.

Limitations and Risks

The demonstration used a fixed maximum sequence length of 200 with padding, which can discard information from longer reviews and introduce padding artifacts. The vocabulary was capped at 8,000 tokens, so rare words were mapped to an unknown token. Training continued for 40 epochs without early stopping, which allowed clear overfitting in the baseline. Training on a single split without extensive hyperparameter search limits claims of optimality. The model also has no explicit mechanism for handling sarcasm or domain shift beyond the IMDB distribution.

Ethical and Responsible Use

Sentiment models trained on movie reviews can embed biases present in the source text (for example, gender or cultural stereotypes in reviewer language). Deploying such a model to moderate user comments or product reviews without additional fairness checks risks amplifying those biases or misclassifying

legitimate criticism. A practical mitigation is to evaluate performance across demographic or topical slices and to combine the model with human review for high-stakes decisions.

Future Improvements

With additional resources, next steps include adding early stopping based on validation loss (the baseline's best point was near epoch 7), testing a bidirectional LSTM or a small Transformer encoder, expanding the vocabulary or using pretrained embeddings, and evaluating on an external review corpus to measure domain shift. Incorporating learning-rate scheduling or weight decay could further improve stability at the higher learning rate.

References

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